PAPER ID-311066

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM V) THEORY EXAMINATION 2024-25
AIR AND NOISE POLLUTION CONTROL

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 07 = 14**

Q no.	Question	CO	Level
a.	Define air pollution.	1	K1
b.	Explain how air pollutants affect animals?	1	K2
c.	Define the term "plume behavior".	2	K1
d.	Define stack height.	2	K1
e.	Define ambient air quality.	3	K1
f.	Differentiate Euro-I and Euro-II.	4	K2
g.	Define noise indices.	5	K1

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 07**

Q no.	Question	CO	Level
a.	Define the composition of Earth's atmosphere and explain the importance of each component in sustaining life on Earth.	1	K2
b.	Define the term "air pollution chemistry" and explain how chemical reactions in the atmosphere contribute to the formation of pollutants.	2	K2
c.	Define ambient air quality and explain the importance of monitoring air quality in urban and industrial environments.	3	K2
d.	Define automotive emissions and explain the primary pollutants emitted by vehicles.	4	K2
e.	Define infrasound, ultrasound, impulsive sound, and sonic boom, and explain how each type of sound is generated and perceived.	5	K2

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

Q no.	Question	CO	Level
a.	Discuss the role of photochemical oxidants in air pollution and explain their formation and environmental impacts.	1	K2
b.	Explain the major global implications of increasing air pollution, focusing on both environmental and human health perspectives.	1	K2

4. Attempt any *one* part of the following: **07 x 1 = 07**

Q no.	Question	CO	Level
a.	Discuss the limitations of the Gaussian Plume Model and its applicability in real-world air pollution scenarios.	2	K2
b.	Define the concept of pollutant dispersion and explain the key factors that determine how pollutants are spread in the atmosphere.	2	K2

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5. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO	Level
a.	Explain the different methods of air sampling and how they are used to measure pollutants such as particulates, gases, and volatile organic compounds.	3	K2
b.	Define wet collectors and explain how they work to remove particulate matter from exhaust gases.	3	K2

6. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO	Level
a.	Define a catalytic converter and explain its role in reducing harmful emissions from vehicle exhaust.	4	K2
b.	Define the Indian emission standards for vehicles and explain how they align with or differ from international standards like Euro-I, Euro-II, and Euro-III.	4	K2

7. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO	Level
a.	Explain how factors like weather conditions, building materials, and ventilation systems affect noise propagation indoors and outdoors.	5	K2
b.	Apply your knowledge of sound measurements to assess the noise levels in an urban environment and recommend measures for controlling noise.	6	K3

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